Sterilization is defined as the process of complete removal or destruction of all forms of microbial life, including vegetative cells and their spores.

4.1 Need for Sterilization

The aim of all sterilization strategies is to kill or remove the unwanted microorganisms. In certain cases, microbes are regarded as potential pathogens and therefore it is essential to eliminate these forms (vegetative and spores) of microbial life. All microbiological techniques require appropriate and adequate sterilization.

Sterilization of culture media, containers and instruments is essential in microbiological work for isolation and maintenance of microorganisms. In surgery and medicine, the sterilization of instruments, drugs and other supplies is important for the prevention of infection.

4.2 Methods of Sterilization

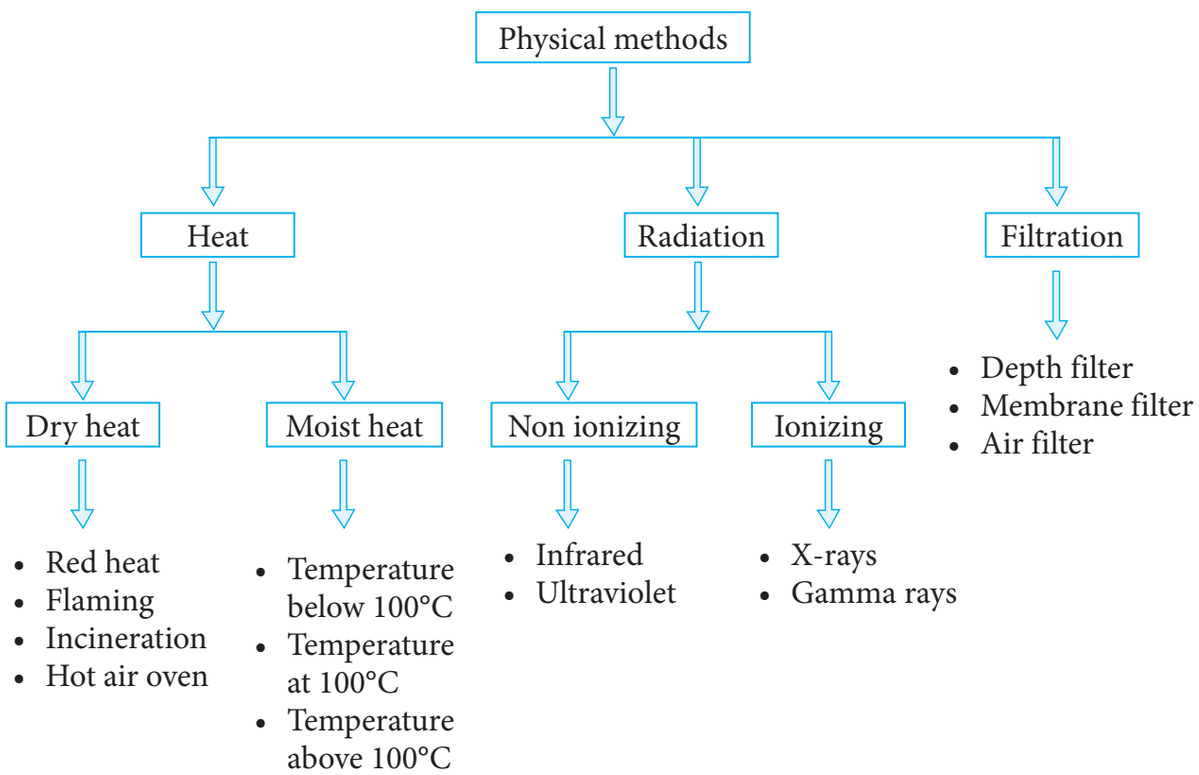
Growth and multiplication of microorganisms can be controlled by removing, killing or inhibiting them using various physical or chemical agents.

4.3 Physical Methods of Sterilization

The various physical methods of sterilization are given in flowchart 4.1

4.4 Sterilization by Heat

Heat is the most rapid and best method of sterilization. It is the method of choice that the material to be sterilized is stable enough to withstand the required temperature necessary to kill the microbes. The time needed for sterilization depends on the initial number of organisms present, type of materials to be sterilized (hence washed and cleaned items are easier to sterilize than dirty ones) and also on the temperature used. Spores need higher temperatures while vegetative bacteria can be destroyed at lower temperatures.



Flowchart 4.1: Physical Methods of Sterilization

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